

End-to-End Computational Pipeline Architecture

Stage 1: Ingestion

- YouTube Data API v3
- YouTube Transcript API
- Local Disk Caching
- Rate-Limit Management



Stage 2: LLM Extraction

- Google Gemini 2.5 Flash
- Pydantic Schema Validation
 - Context Windowing
 - View-Weight Injection



Stage 3: Network Science

- Directed Bipartite $G=(V,E)$
- Centrality Metric Suite
- Louvain Modularity (Q)
- ICM Cascade Simulation



Stage 4: Visualization

- Interactive PyVis / Vis.js
- ForceAtlas2 Physics
- Publication Graphics
- Academic Dissertation